

## Lecture 3 Module 1

Monday, 24 August 2020 9:58 PM

$X, Y \subseteq U$

$X \sim Y$  (there is a bijection between  $X$  &  $Y$ )  
 $X$  is equivalent to  $Y$

$X \leq Y$ :  $X$  is equivalent to a subset of  $Y$ , ' $Y$  **dominates**  $X$ '



$\Leftrightarrow$  There is a one-one  $f: X \rightarrow Y$ .

- what kind of a relation is ' $\leq$ '?

$\leq \subseteq P(U) \times P(U)$

- reflexive, transitive

- antisymmetric? i.e.

$X \leq Y, Y \leq X \Rightarrow X = Y$

Schröder-Bernstein Th:  $X \sim Y$

$X < Y$ :  $X \leq Y$ , but  $Y \not\leq X$  i.e.  $X \not\sim Y$   
**strict domination**

Obs.  $X \leq Y, Y < Z \Rightarrow X < Z$ .

$\omega$ : set of natural numbers. ( $\mathbb{N}$ )

- For any set  $X$ , define its successor

$X^+ := X \cup \{X\}$ .

-  $\emptyset$ ;  $\emptyset^+ = \{\emptyset\}$ ,  $\emptyset^{++} = \emptyset^+ \cup \{\emptyset^+\}$   
 $\{ \emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\} \}$

$$\begin{array}{ccc}
 \downarrow & \downarrow & \downarrow \\
 0 & 1 = \{0\} & 2 = \{0, 1\} \dots
 \end{array}
 \begin{array}{l}
 = \{\emptyset\} \cup \{\{\emptyset\}\} \\
 = \{\emptyset, \{\emptyset\}\}
 \end{array}$$

$$- \omega := \{0, 1, 2, \dots\}, n \in \omega$$

## Finite & Infinite Sets

-  $X$  is finite iff  $X \sim n$ , for some  $n \in \omega$ .

-  $X$  is infinite, otherwise,  $|X| = n$ : cardi-  
nality  
 $|X|$

## Consequences

•  $\omega$  is infinite

•  $\omega \leq X \Rightarrow X$  is infinite

•  $X$  is infinite  $\Rightarrow \omega \leq X$  (AC)

$X \neq \emptyset$ .  $x_0 \in X$ .  $X \setminus \{x_0\} \neq \emptyset$

$x_1 \in X \setminus \{x_0\}$ .  $X \setminus \{x_0, x_1\} \neq \emptyset$ .  $x_2 \in X \setminus \{x_0, x_1, x_2\}$

•  $X$  is infinite  $\Leftrightarrow$  there is  $\gamma$ :

$$\gamma \subsetneq X \text{ and } \gamma \sim X$$

- Dedekind's defn

$$(\omega)$$

$$X \setminus \gamma$$

- Ex.



Proof  $(\Leftarrow)$ :  $X$  finite

$(\Rightarrow)$   $\omega \leq X$   $f: \omega \rightarrow X$  one-one.

Define  $h: X \rightarrow X$

$$h(n) = \begin{cases} f(n+1) & \text{if } n = f(n) \\ n & \text{if } n \in X \setminus f(\omega) \end{cases}$$

$h$  is one-one &  $h(\underbrace{X}_{\neq \emptyset}) \subsetneq X$ .